

Electricity theft: Overview, issues, prevention and a smart meter based approach to control theft

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ABSTRACT

Non-technical loss (NTL) during transmission of electrical energy is a major problem in developing countries and it has been very difficult for the utility companies to detect and fight the people responsible for theft. Electricity theft forms a major chunk of NTL. These losses affect quality of supply, increase load on the generating station, and affect tariff imposed on genuine customers. This paper discusses the factors that influence the consumers to steal electricity. In view of these ill effects, various methods for detection and estimation of the theft are discussed. This paper proposes an architectural design of smart meter, external control station, harmonic generator, and filter circuit. Motivation of this work is to deject illegal consumers, and conserve and effectively utilize energy. As well, smart meters are designed to provide data of various parameters related to instantaneous power consumption. NTL in the distribution feeder is computed by external control station from the sending end information of the distribution feeder. If a considerable amount of NTL is detected, harmonic generator is operated at that feeder for introducing additional harmonic component for destroying appliances of the illegal consumers. For illustration, cost-benefit analysis for implementation of the proposed system in India is presented.

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1. Introduction

Generation, transmission and distribution of electrical energy involve many operational losses. Whereas, losses concerned to generation can be technically defined, but Transmission and Distribution (T&D) losses cannot be precisely quantified from the sending end information. This illustrates the involvement of some non-technical parameters in T&D of electrical energy. Technical losses occur naturally and are caused because of power dissipation in transmission lines, transformers, and other power system components. Technical losses in T&D are computed from the information about total load on the grid and total energy billed (Oliveira and Barioni, 2009). NTL cannot be precisely computed, but can be estimated. In general, NTL are caused by the factors external to the power system. So, it is very difficult to represent electricity theft as a function of actions responsible for the same. In many developing countries, NTL account to about 10–40% of their total generation capacity (Smith, 2003). Electricity theft forms a major chunk of the NTL. It includes illegal tapping of electricity from the feeder, bypassing the energy meter, tampering with the energy meter and several physical methods to evade payment to the utility company (Dick, 1995). Of which, illegal tapping of electricity and tampering with energy meter are the most identified and accounted

ways of theft. Electricity theft can also be defined as using electricity from utility company without a contract or valid obligation to alter its measurement (Smith, 2003). Electricity theft is a complex problem with many parameters to be evaluated before implementing any measures to detect and control that. These parameters include some issues like social, economical, regional, managerial, political, infrastructural, literacy rate, criminal, mafia, corruption, effect on genuine customers, power quality, safety, and uncertainty in time period during which electricity is stolen. Apart from illustrated methods, billing irregularities is another form of NTL in many developing countries. Corruption and lack of commitment in employees of utility companies to control the illegal consumption are the reasons for billing irregularities (Gulati). For instance, an employee might not read the correct value of the total energy displayed on energy meter or may intentionally give lower value than the exactly read one (Gulati). Few household and industrial customers are also responsible for nonpayment of utility bills because of their weak economic situation.

Fig. 1 illustrates the total installed generation capacity and Fig. 2 shows overall T&D losses recorded in several countries. Worldwide T&D losses are more than total installed generation capacity of Germany, UK and France. It is estimated that utility companies worldwide, lose more than \$25 billion every year due to electricity theft, of which utility companies in India loses about \$4.5 billion every year (Overview of power distribution). Recovery of about 10% of which could conserve about 83 000 GWh of electric energy and reduce carbon dioxide emissions by 9.2 million tons annually (Pyasi and Verma, 2008). Losing about \$4.5 billion every year,

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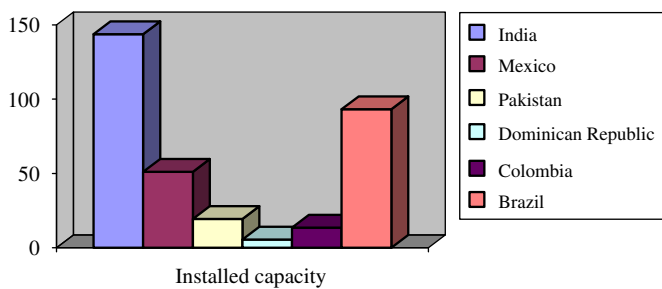


Fig. 1. Total installed generation capacity in million kW in the year 2006 (International Energy Annual, 2006a).

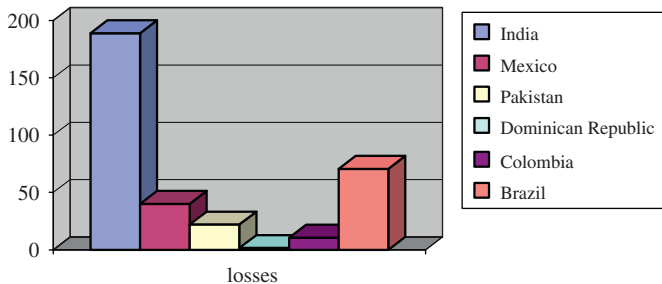


Fig. 2. Overall T&D losses in billion kWh in the year 2006 (International Energy Annual, 2006b).



Fig. 3. A distribution feeder with numerous interconnections (Keulenaer, 2007).

which might be approximately equal to or more than their overall profit, illustrates these utility companies are supplying power at no profit and even in financial losses. This situation might drive these utility companies out of business and poses threat to their survival. It is an inevitable situation that, utilities have control the theft to continue in the business, it is important to design a mechanism that reduce NTL.

Fig. 3 displays a distribution feeder with many interconnections, which confirms our concern about complexity of the problem displaying various illegally fed loads tapping by tapping electricity bypassing their energy meter. Several studies show that, at certain points of time, NTL are very high compared to the value estimated by utility companies.

In fact, not much technical work has been done in detecting and controlling electricity pilfering. However, several non-technical measures have been carried out. For instance, in India, adoption of several awareness programs and initiatives by the government have brought down the overall T&D losses from 38.86% in 2001–2002 to 34.54% in

2005–2006 (Overview of power distribution). As seen, although there is some decrement in the losses, electricity distribution system still needs to be reformed for proper estimation and reduction of electricity theft. In light of the situation, an appropriate solution that can detect the existence of theft as well as help the utility companies to punish the illegal consumers would be highly appropriate. Replacing conventional energy meters with tamper-proof smart meters would be the initial step in this process.

2. Effects of electricity theft

Negative effects of electricity theft are severe and dangerous. Primarily, electricity theft affects the utility company and then its customers. Electricity theft overloads the generation unit. In addition, quality of electricity supply is adversely affected, as the utility company has no estimate about the quantity of electricity to be supplied to genuine customers as well as illegal consumers. This overload might result in over voltage, affects the performance and even damage appliances of customers. This huge amount of NTL might trip the generation unit, which interrupts power supply to all customers (Sullivan, 2002). This unpredictable amount of additional load may lead to brownouts and blackouts during the peak load period. In order to maintain good power factor and flat voltage profile along the feeders, sufficient reactive power has to be supplied besides the supplied electricity. Load shedding should also be done to compensate the voltage collapse during the peak load period. In addition, VAR compensation is very difficult without complete information about the total load flow because of the theft (Begovic et al., 2000).

In energy market, utility companies expect to earn their money back from the customers for the service provided, but, most of this is lost due to the NTL. Electricity theft is a serious concern for utility companies as they are under threat of survival because of these incurring economic losses. Evidently, some utility companies in developing countries lose a considerable amount of their total revenue. These economic losses affect the utility company's interest in development of the devices to control these losses. In addition, utility companies are forced to impose excessive tariff on genuine customers as they cannot afford all these losses by themselves which is absolutely not ethical to make genuine customers and utility companies responsible for the energy consumed by illegal consumers. In addition, illegal tapping of electricity raises safety concerns like electric shocks and even the death of a person who operates it. Improper handling of the distribution feeder might pose danger to the whole community, as these wires might start sparking and may cause fire during extreme weather conditions. Illegal consumers initiate tapping electricity from distribution feeders during scheduled power cuts.

3. Methods of stealing electricity

As illustrated earlier, the most common forms of stealing electricity are tapping electricity directly from the distribution feeder and tampering with the energy meter. Tampering with energy meters is done to manipulate the meter reading. Tampering with the electromagnetic meters include interruption of rotating disc by inserting a film or depositing a high viscous fluid, or using strong magnets like neodymium magnets. Apart from these techniques, radio frequency devices can be employed to tamper the electronic energy meters. Other methods of tampering energy meter include, inter-changing the incoming and outgoing terminals of the meter, damaging the pressure coil of the meter, resetting the reading, unwanted harmonics (KSEB Officers' Association) and exposing the meter to mechanical shock. Other

important engineered ways of tampering with the meter without damaging them are illustrated in the later portion of the section.

Current transformers (CT) are employed in energy meters those measure energy consumption of large loads to match the current rating of grid with the energy meter. Secondary side wires of CTs are generally insulated, but this insulation is damaged before tapping energy from either one or both the wires. Tapping electricity from these wires cause the meter to read less current or even zero current based on the number of the wires tampered. Altering the quantity of current in the breaking wires can directly affect the energy meter reading. In addition, exchanging the position of breaking wires shift the phase and also alters the meter reading. Apart from these methods, voltage can also be tapped from wires of the energy meter, which makes it read lesser quantity than original consumption. Improper calibration and illegal de-calibration of energy meters can also cause NTL (Cespedes et al., 1983). Two-watt hour meters employed for reading the energy consumed by large loads with three phase electric supply are tampered in the following way: first, terminal seal is damaged, one of the terminals of the meter is connected to load and the other terminal of the meter is left open, second terminal of the load is grounded making the meter assume that there is no energy consumption from the grid. In three phase meters, phases can also be shifted to lower the power consumption reading by the energy meter.

One popular way of lowering the energy meter reading without directly tampering the meter is shown in Fig. 4. In this case, illegal consumers use just one of the three phases, disconnects neutral from the distribution feeder, and use a separate neutral for return path. Here, energy meter assumes that the voltage between connected phase and this new neutral is zero, implying the total energy consumed to be zero. Another engineered way of stealing electricity is, isolate the neutral and disturb the electronic reference point by physically damaging the meter, and then the voltage to be read by energy meter can be manipulated by controlling the neutral.

In developing countries, bypassing the meter and tapping electricity directly from the feeder are done only at desired hours of the day. It is a common practice that such consumers use electricity from the utility grid through the energy meter during the time when their consumption will be low and steal electricity during the time when their consumption will be high. Such consumers use legal electricity for small household loads and illegally tapped electricity for heavy loads like water pumps and air conditioners, etc. Illegal consumers tend to pilfer electricity

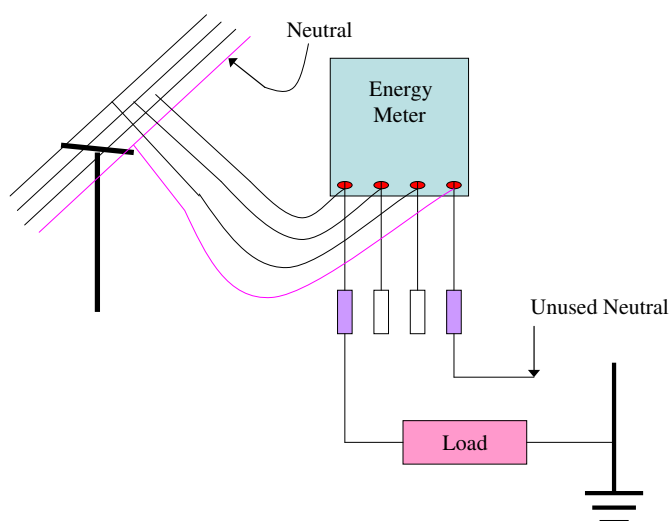


Fig. 4. Regulating the supply voltage to manipulate the meter reading (Power theft: a blow on customer).

Table 1

General tariff chart for electricity consumption for residential customers in Andhra Pradesh, India under slab system

Energy consumption in kWh/month	Tariff per 1 kWh in US \$
0–99	0.0444
100–199	0.0533
200–299	0.0664
300–999	0.0886

when the probability of inspection by the utility employees is comparatively low. This form of theft is very difficult to quantize, as the energy consumption pattern is not uniform over a period of time. Another common practice is that, people tap a portion of electricity illegally during event gatherings and public functions. Sudden inspection of overhead distribution feeders supplying customers with suspicious load profiles might yield good results in detecting illegal tapping. Corrupt staff of the utility companies might take bribes from illegal consumers to help them allow such practices. In addition, corrupt employees are partially responsible for billing irregularities, as they can record quantity lower than the original energy consumption. Table 1 illustrates the approximate tariff plan of a residential household.

Consider the overall energy consumption of a particular customer is about 210 kWh in a month. Corrupt employees might record that as 190–199 kWh to give a lower utility bill to a specific customer. This might lead to huge economic losses and also poses a difficult challenge for utility companies in maintaining good power quality. These customers pay for that unbilled energy in their future billing cycles, but this confuses the utility company about the quantity of accounted and billed energy. Sometimes, feeding bribes for line technicians, electricians and maintenance staff for rectification of technical faults makes people opt for illegal electricity. Beneficiaries of such corruption include illegal consumers, utility managers, line technicians, distribution and billing staff. Total money bribed to the utility employees is less compared to the money that they pay for consuming the same amount of electricity legally. Apart from these methods, sophisticated technologies like remote sensing devices and the crystal frequency of integrated circuits is tampered. Illegal consumers always have answers for most of the challenges posed by utility companies through implementation of new technologies.

4. Factors that influence illegal consumers

There are few factors that encourage people to steal electricity. Of which socio-economic factors influence people to a great extent in stealing electricity, a few of them are:

- A common notion in illegal consumers is that, it is dishonest to steal something from their neighbor but not from the state or public owned utility company.
- Growing unemployment rate and bad economic situation of the customer.
- Lower illiteracy rate of consumers, as they might not be aware of the issues, laws and offenses related to the theft.
- In view of economic conditions of the customer, quantity of electricity theft is proportional to the tariff of electricity consumption in that region.
- Countries with weak enforcement of law against electricity theft record high proportion of theft.
- Corrupt politicians and employees of utility companies responsible for billing irregularities.

Table 2
Tariff charged for electricity consumption in several countries

Country	Tariff in cent US \$ per 1 kWh	Information as of year
Australia	7.11 (Power Engineering International)	2006–2007
Canada	6.18 (Power Engineering International)	2006–2007
Denmark	42.89 (Power Engineering International)	2006–2007
Finland	6.95 (Power Engineering International)	2006–2007
Germany	30.66 (Europe's Energy Portal-Prices, Statistics, Supply & Security)	2009
Italy	37.23 (Europe's Energy Portal-Prices, Statistics, Supply & Security)	2009
Malaysia	7.42 (Suruhanjaya Tenaga, 2007)	2007
South Africa	10.15 (ESKOM)	2008–2009
Sweden	27.34 (Europe's Energy Portal-Prices, Statistics, Supply & Security)	2009
UK	18.59 (Europe's Energy Portal-Prices, Statistics, Supply & Security)	2009
USA	9.28 (Power Engineering International)	2007

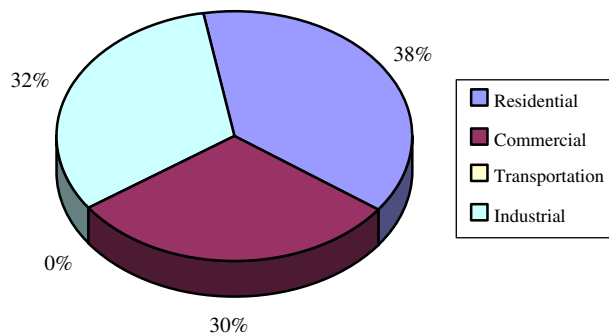


Fig. 5. End use electricity consumption by sector in USA in the year 2004 (US Energy Information Administration, Department of Energy).

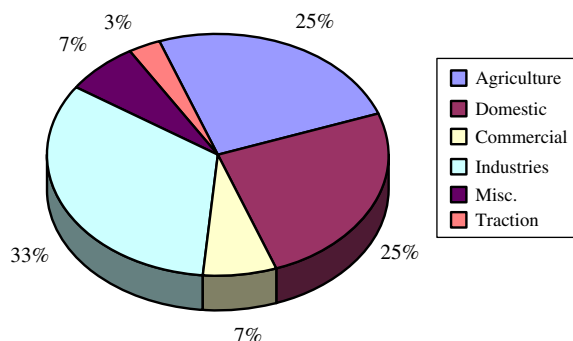


Fig. 6. End use electricity consumption by sector in India in the year 2004 (Energy, Resources & Environmental Policies, Bryn Mawr college).

- Higher energy prices deject some consumers from buying electricity. Table 2 illustrates tariff on electricity consumption in different countries. In light of this high tariff rates, some citizens of rich and highly educated communities might steal electricity to avoid payment of huge utility bills (*Challenges in the Electricity Sector: Jamaica at the Crossroads*).
- Lack of electricity supply to the neighborhood would be a reason for theft in slums and poor communities.

In addition, probability that an illegal consumer steals electricity from any distribution feeder depends on the sector where the feeder is located. Fig. 5 shows the pattern of electricity consumption by sector in USA, where, industrial sector forms a major chunk of the electricity generated. In developed countries, because of the transparency in distribution, electricity theft might not be possible in industries and commercial complexes. Whereas, in developing countries, agriculture forms a major chunk of total energy consumption following the industries, as shown in Fig. 6. Due to the

lack of proper metering infrastructure in agriculture and related sectors, considerable amount of electricity is being stolen here.

5. Cases and issues

Electricity theft is a crime, sin, offense, or still an ignorable mistake in different countries depending on the local parameters. In this section, several cases of electricity theft are illustrated in detail. In India, electricity consumption for agriculture is provided at free of cost in several states, as several political parties promise such incentives which prevent utility companies from collecting their bill. Here, illegal consumers tap electricity from a location on a distribution feeder close to agricultural equipment, which overload the nearby transformer. Major cities like Mumbai and Delhi account for very high NTL of about 30% (*Business News, BBC*), where some corporate companies owe pending payments to the utility company. In addition, there is an interesting case in which, mirror investigation is carried out on power mafia operations in slums of Mumbai. Here, this mafia is powerful, comprising of political workers, spies, informers, and other workers to maintain new and existing connections. These people are responsible for coordination of customers and utility staff. They steal electricity from distribution feeder in the neighborhood using their own cables, which is then transmitted and distributed using inter connecting cables. These people hide their equipment to avoid inspection by utility company in neighborhood. Getting an illegal connection from this mafia is very easy and cost effective compared to a regular connection from a utility company. They allow unlimited usage electricity at a flat rate of about \$5 per month compared to an average bill of about \$15 (*Brihan Mumbai Electric Supply & Transport Undertaking Tariff Booklet*) for a family of four, if billed by the utility company. Poor people with low monthly income who cannot afford to pay for genuine electricity prefers such electricity connections. In view of these irregularities, utility companies still need to collect about \$2 billion as pending bills (*Mumbai Mirror*).

In Canada, most often, motivation to steal electricity is a temptation not to pay for the electricity consumed, or to pay less. This is an obvious situation, people who process large amounts of marijuana steal electricity, as the consumption would be very high (*Cannabis News*). This is similar in the USA, people who cultivate marijuana illegally, steal electricity to hide their overall electricity consumption to avoid police inspection and prosecution. Electricity theft has become dominant during the 2008–2009 economic recession and financial suppression. In Pakistan, electricity theft is a common practice in poor residential areas where there is no access to a distribution feeder. This is called Kunda system which is accepted by the employees of utility company as a truth of life in such poor neighborhoods without adequate infrastructure. It is discovered that some government agencies including the Army

owes large amounts debts to the utility company. In 2000, Karachi electric supply corporation (KESC) reported that only 52% of its 1.67 million customers were paying their electricity bills (Smith, 2003). Here are more cases that evaluate the discussion are:

- In Malaysia, inspection teams revealed that 587 out of 684 suspected cases are confirmed to have stolen electricity. Here, utility companies proposed to fix energy meters for illegal consumers at a low flat rate.
- In Mexico, illegal customers called little devils, overload the electrical grid and affect many residential and commercial loads. It is estimated that utility companies lose about \$475 million revenue every year (News and Trends: Latin America, Alexander's Gas & Oil Connections).
- In Armenia, South Africa, about 80–90% of residential customers evaded their payment. Nonpayment is not confined to only poor communities, but also to powerful people who are aware that their subscription will not be disturbed irrespective of their payment. In Soweto, about six tons of cable used for illegal tapping was seized in a span of six months (Smith, 2003).
- In Indonesia, Perusahaan Listrik Negara (PLN), local utility company, claimed that there are about \$16.85 million to be collected from unpaid cases, of which the military owe about \$3.1 million.
- In Jamaica, illegal tapping of electricity is common in inner city communities and tampering with the meter is common in rich neighborhoods and commercial sectors (Steadman).
- In Brazil, analysis was done to analyze the relation among non-technical losses and socio-economic conditions, which illustrated that, neighborhoods with high crime rate and low income and have higher chances of electricity theft.

6. Techniques for estimation and control of electricity theft

In many developing countries, electricity theft is publicly visible at many distribution feeders. In order to control electricity theft, several technical as well as non-technical methods are implemented for estimating and controlling theft. Though detection of point of theft is important, it is not a solution to control the theft. Some non-technical measures include, subsidies to customers with low income in the form of lower tariff might make them feel that electricity is affordable and encourages them to use genuine form of electricity. Also, efficient business practices and proper audit of the electricity consumption at distribution level deject illegal consumers from manipulating the energy meters. In addition, electricity theft has to be considered as a serious offense and illegal usage of any form of energy belonging to the utility company should be made punishable under the court of law. Laws and policies have to be strictly enforced such that political leaders do not protect corrupt employees and illegal consumers responsible for theft. Researchers have proposed and developed several techniques for detection and estimation of electricity theft, of which, a few methods are illustrated in this section.

Bandim et al. (2003) proposed utilization of a central observer meter at secondary terminals of distribution transformer. Value of energy read by the central observer meter is compared with the sum of energy consumption values read by all energy meters in range. These two values of current are compared to estimate the total quantity of electricity that is being consumed illegally. Vigilant energy metering system (VEMS) (Anand and Naveen, 2003) is an advanced energy metering system that can fight electricity theft. It identifies probable locations of theft and helps the utility companies to control theft. Nagi et al. (2009) proposed a novel approach of using genetic algorithm-support vector machines (GA-SVM) in detecting electricity theft. Load consumption data of all customers

is collected and data mining techniques are used to filter and group these customers based on their consumption patterns. Customers are grouped into different classes based on the extent of the abnormality in load profile and then the customers with high probability of theft are inspected. In addition, Nizar and Dong (2009) used extreme learning machine approach to evaluate abnormal load behavior indicating electricity theft based on a load-profile evaluation. These methods are carried out before inspecting the illegal consumers, based on the energy consumption data collected at proper intervals of time.

Power line impedance technique determines the location of an illegal consumer with reference to a genuine customer on the same distribution feeder. Impedance of a power line is measured at 50 Hz, and is then with a signal of 2 V at 150 kHz signal is sent. These two impedances are compared to calculate the location of theft with respect to the location of a genuine customer. An illegal connection is concluded to be located between two genuine meters, if the illegal consumer's load is directly connected to distribution feeder or the illegal consumer's load will be parallel to a genuine customer if the thief uses extra phase before a genuine customer (Pasdar and Mirzakuchaki, 2007). Periodic inspection of the illegal connections involves huge labor and strain for vigilance officials. The shunts detecting equipment is time efficient and helps detection of electricity theft in underground and inaccessible distribution cables. This device can reduce the average time period for inspection and detection of a neighborhood from about two days (48 h) to 4 h (Parra and Calderon, 2006). Venaas and Becker (1980) patented an energy meter which has an indicator light that glows if the installed meter indicating is dislocated. In case if the energy meter is not properly calibrated by the utility company, a linear model can be used for losses evaluation, as, most of the customer's average current consumption is between 10% and 30% of nominal current (Cespedes et al., 1983).

A smart meter is an advanced energy meter that identifies energy consumption in more detail compared to a conventional energy meter. In addition smart meter can reliably and securely communicate the data back to the local utility for monitoring and billing purposes. Smart meters with great networking capability and advanced software tools are very difficult to tamper and hack. Smart meters with passwords are still difficult to tamper and promise a great deal of implementation in future smart grids (McLaughlin et al.). Zeng et al. (2009) proposed a new method to monitor the street lamp power cables and theft of cables. In this method, resonant frequency and equivalent capacitance of the cable are calculated by sending a frequency varying current signal to the street lamp power system. These values are used to detect the theft of cable, and then the location of theft. A microcontroller based energy meter proposed by Jamil et al. (2004) facilitates the utility company to monitor and control the power supply to its spatially distributed consumers. This meter acts as a check meter to detect the meter tampering. However, e-metering systems can collect and process data, and can detect abnormalities in load profiles indicating electricity theft (De et al., 2003).

Revenue Assurance and Audit Process (RAAP) is composed of macro-functions those detect and analyze revenues involved in electricity theft. Also, Mano et al. (2004) suggests proper design and implementation of rules for investigation of illegal consumers. RAAP is targeted at improving revenues for utility companies by reducing commercial losses by about 20% each year. Perez et al. (2005) suggest that proper implementation of strategies in deployment and maintenance of the distribution networks can control commercial losses. In addition, strengthening the transformers at substations with higher configuration ones is also suggested. These suggestions can also be implemented in other African countries with the similar situations in order to improve the efficiency in T&D.

For instance, in India, Electricity Act 2003 has made electricity theft a punishable offense and gave freedom to vigilance officials to inspect and detect electricity theft. In Pakistan, KESC has obtained a fatwa, a decree, from Islamic scholars, which declared that electricity theft is a sin (News Article, The Dawn Media Group). Good strategy for fighting corruption has to be developed, considering the political scenario, business processes, management techniques, and technologies in metering and distribution monitoring based on the location. Apart from empowering the laws and campaigning against the theft, regularization of agricultural connections are to be enhanced. All contracts for deployment and maintenance of the transmission and distribution sector are to be outsourced, based on the performance of the enterprise to which the bid will be awarded. In addition, all the norms and standards are to be strengthened such that, people must fear to make an attempt to corrupt the system or steal electricity. Regular audits and local searches have to be scheduled so that the illegal tapping can be controlled to seize these offenders (Gulati). GE has patented a smart energy meter that read consumption of electricity correctly even if the in-going and out-coming meter terminals of the meter are reversed. This invention stopped people from using their energy meter in reverse direction to reduce their utility bill.

7. Setbacks for implementation of available measures

Major setbacks in implementation of existing policies and regulations for controlling electricity theft include weak infrastructure, corruption, and inefficiency of the staff of utility companies. The utility company as a unit has rules and guidelines to control electricity theft, but corrupt employees in the same system display least dedication towards implementation of the measures to reduce theft. In most of the tampering cases, damaged meter terminals and seals/insulations are not visible for inspection, which is also a major setback. Illegal tapping of electricity from distribution feeder and tampering with the existing energy meter can be battled only with public participation, which is impossible. Total losses incurred by utility companies because of theft are very large and may run them into bankruptcy, which makes it difficult to adapt and implement measures at the cost of few billion dollars. In addition, even the priorities in investment for implementation of measures to reduce theft, might also be prone to corruption (Depuru et al., 2010a, b). These concerns warn us about formulating a strategy for scheduling the implementation, as the investment by utility companies has to be collected back in a few years.

8. A conceptual design for controlling electricity theft

Our proposed approach is a paradigm shift from the conventional method of identifying the illegal consumer, by physical observation of the distribution feeder or evaluation of load pattern of all customers (Depuru et al., 2010a, b). Fig. 7 shows flow chart of operation for the proposed system. Under normal operation, smart meters in all households transmit instantaneous values of power consumption, power factor and current to the external control station (ECS) located at the utility company. ECS computes the total non-technical losses from these received values. If the computed non-technical losses are more than 5% of the distributed energy, then the ECS will send a control signal to the internal control station (ICS) of smart meter to break the electric supply to the genuine customers. In this process, primarily, the genuine customers are identified and are isolated from the electric grid, leaving the illegal consumers continue to draw energy from the grid. Then the harmonic generator is switched on for few seconds and is switched off after that period just before the scheduled power cut for that

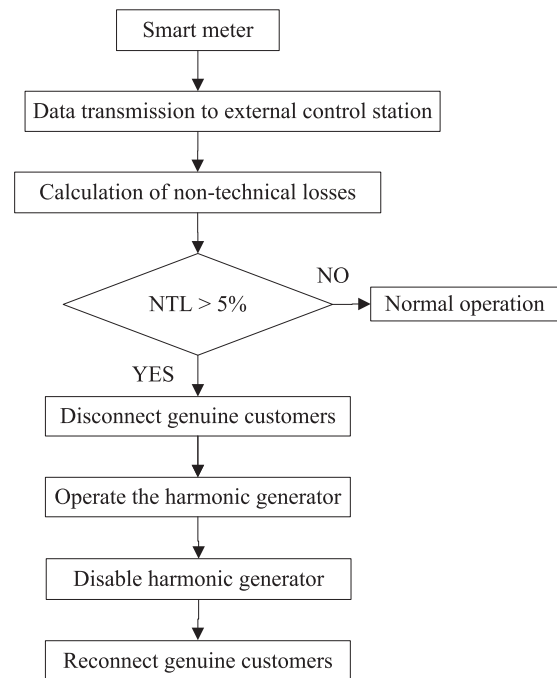


Fig. 7. Flow chart of the proposed system.

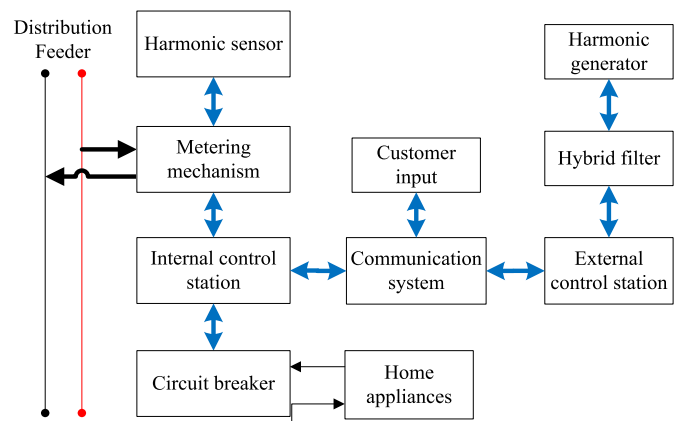


Fig. 8. Proposed model of the smart meter.

neighborhood. Here, purpose of operation of harmonic generator is to introduce unwanted harmonic component into the distribution feeder which would affect the performance of home appliances of illegal consumers. After operating the harmonic generator, ECS signals the ICS of all smart meters to restore the power supply back to genuine customers, after switching off the harmonic generator.

8.1. System design

As shown in Fig. 8, the proposed smart meter architecture has an energy meter, ICS, circuit breaker, harmonic sensor, and a communication system. ICS is the central processing unit of smart meter, which controls all components of the smart meter. Energy meter portion of the smart meter measures the amount of electricity consumed by the household. Circuit breaker is used to disable electricity supply to genuine customers after receiving a request from the ICS. Harmonic sensor is used to detect existence of additional harmonic component of the current over the THD, in electricity supplied by utility company. This additional value of the harmonic component is calculated from the difference between the

total phase current from the distribution feeder and its fundamental component of the current based on instantaneous reactive power theory (Liu et al., 1999). Data transfer between smart meter and ECS over the communication medium has to be secure, reliable, and efficient. Therefore, such communication mechanism should satisfy standards regarding radio communication protocols, networking interfaces and industry standard security specifications.

ECS will be designed to have a workstation with a wireless transceiver and control system. This workstation is used to compute total non-technical losses and issues control signals to ICS of all smart meters accordingly. Wireless transceiver is used to send and receive data and control signals among the smart meters, ECS, and ICS. ECS is capable of managing and controlling the supply of electricity to any or all home appliances. Conceptual model of a harmonic generator can be designed by using four thyristors and a pulse wave generator for firing the gate of thyristors. The resultant model will introduce additional harmonic component in the current. Hybrid filter is connected between distribution transformer and harmonic generator to prevent the damage of the power system components from unwanted harmonic component. Hybrid filter consists of an active filter and a passive filter. Active filter can be built using four fully controllable IGBTs, whereas, the passive filter is a combination of inductors and capacitors.

8.2. Design issues and constraints

Certain design issues and constraints are to be considered while designing the proposed system. Some of these issues and constraints are tapping of electricity from the feeder is assisted by poor infrastructure at the metering and distribution level. In most of the distribution feeders, there will be about 50–100 interconnections on each pole those support the overhead distribution feeder as illustrated in Fig. 3. This makes it difficult to identify illegal consumers even after identifying the location of theft on distribution feeder.

In this context, harmonic generator has to be designed keeping in view of the acceptable range of skin effect resistance and THD in the transmission feeder that might be caused by additional harmonic component. In addition, harmonic filter used to protect the power system components from the additional harmonics, should be designed considering the overall impedance of the ac system at the harmonic frequencies. Operation of the proposed system might have a setback in the form of strong opposition from the neighborhood in the process of avoiding the acceptance of theft. Thus, genuine customers have to be educated about effect of electricity theft on their utility bills. Since all the household appliances are designed to operate at fundamental frequency, introduction of harmonics into the distribution feeder may yield to their unsatisfactory operation. Therefore, the proposed system is designed to chastise the appliances of illegal consumers; however, the appliances of genuine customers are protected.

8.3. Advantages

The proposed system is a reliable and non-tamperable alternative to the conventional energy metering system. While coming to the specific advantages of the system, harmonic sensor attached to the smart meter protects the appliances of genuine customers both from the effect of induced harmonic content and from the transients caused by any natural calamity. In addition, the proposed system results in cost savings for both the customer and the utility company in its long term operation. For instance, in India, utility companies supply electricity to about 150 million customers including household, commercial and industrial customers, of which about 20 million (Electricity sector in India) might not have a genuine metered connection. So, if the whole system has to be revamped, about 130 million energy meters have to be

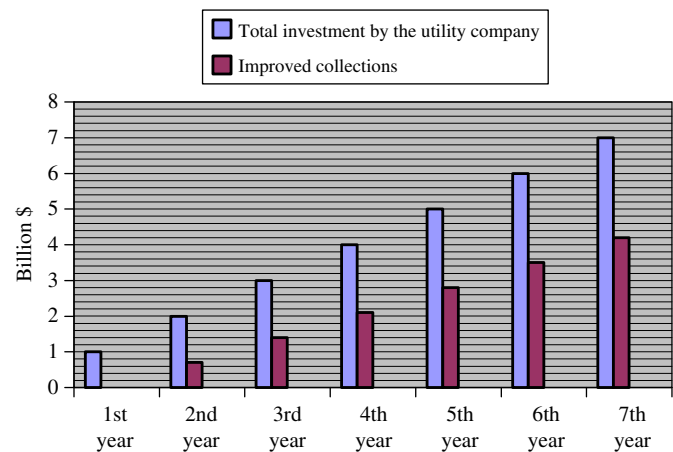


Fig. 9. Expected capital cost and savings graph.

attached with the auxiliary unit to perform the operations in addition to the regular metering, and for about 20 million customers, both the energy meters and the auxiliary units have to be installed. If the auxiliary unit is estimated to cost about \$40 per each unit; auxiliary unit and the energy meter at \$60 for each unit, total investment would be about \$6.4 billion. This huge budget might initiate formulation of a strategy for prioritizing locations for installation, operation and investing that money in several steps. We estimate that, if \$1 billion is invested on this project every year, whole power distribution system in India can be upgraded in seven years as shown in Fig. 9. Considering the current scenario of economic losses incurred by the utility companies due to electricity theft, this infrastructural upgrade and installation of new smart meters has a potential to save about \$4 billion every year. Return on the investment is obtained in the form of improved collections in the utility bills from the new customers, who were previously illegal consumers. Once the up-gradation of the distribution system is started, many irregular and illegal electricity connections are regularized, indicating that several people who were illegal consumers will be genuine customers. So, return on the investment can be expected from the second year after initiation of the project. This improvement in the revenue of utility companies can enhance the country's economy. As illustrated in cost–benefit analysis, return on total capital invested can be obtained in a few years. As this upgrade would help the utility company in many other ways than just reducing the theft, utility companies might adapt this system with minor alterations satisfying the local requirements and parameters. Smart meter is non-tamperable, it would be very difficult for the illegal consumers to install the harmonic filter to their smart meter system and such filters are very costly. It would be cost efficient option for the illegal consumer to pay the utility company than trying to install a filter or to pilfer electricity. In addition, this system can be implemented anywhere in the world with similar economic, political and regional conditions. The main constraint might be incompatibility of smart meter with local situations and policies. Harmonic generator and hybrid filter should be designed based on the parameters such as frequency, THD and ac resistance of the feeder should be designed based on the local parameters.

8.4. Liability issues

The fear that illegal consumers will be punished regardless of the quantity of theft they were responsible, might bring the required change in their view about illegal consumption. For instance in India, any vigilance official from law enforcement agencies can break-open and inspect any premises where the electricity has been or is suspected to be used illegally (Offences and Penalties—Theft of Electricity). Evidently from a case in Andhra Pradesh, India,

enforcement of the new antitheft law and implementation some non-technical measures like neighborhood checks and education about electricity theft, reduced total NTL from 38% in 1999 to 26% in 2003 after detection after regularization of 2.25 million unauthorized subscriptions. They can also search, seize and remove equipment that has been or likely to be used for illegal consumption of electricity. Bhatia and Gulati (2004) suggested some modifications to the legal framework while removing legal setbacks and further empower enforcement-authorities to control the theft.

In view of the severity of the problem, the existing constitution for punishing the illegal consumers has to be revamped such that, they will be punished directly. Damaging the appliances of illegal consumers responsible for theft would be a valid portion of solution to the problem. As mentioned earlier, all the consumers will be educated about the operation of harmonic generator and possible damage to their equipment, while installing smart meters in a neighborhood. After educating the people, ample time must be given for the illegal consumers to voluntarily regularize their subscription. If electricity theft exists even after such warning, then illegal consumers have to pay for their offense. The proposed system has to be implemented in a test environment to analyze the effect of introduced harmonic component on the consumer's equipment. So, the proposed harmonic generator would be a model for demonstration, which creates a fear that, if harmonic generator is operated in their location and if they were using electricity illegally, then there is an acute chance that their devices might get damaged. However, the filter and other protection equipment are thoroughly designed for reducing effect of the additional harmonic component on the equipment of the utility company, damage cannot be completely avoided. So, harmonic generation units are initially deployed in a few locations to further penetrate the fear of using electricity illegally. This system can be implemented in any country, provided their electricity regulations, policies and laws allow implementation of such chastising devices.

9. Conclusion

This paper defines electricity theft in social, economical, regional, political, literacy, criminal, and corruption points of view. This paper illustrates various cases, issues and setbacks in the design, development, deployment, operation, and maintenance of electricity theft controlling devices. In addition, various factors that influence people to steal electricity are discussed. This paper illustrates the effect of NTL on quality of supply, burden on the generating station and tariff imposed on genuine customer. In addition, we proposed a system to detect and reduce the electricity theft by chastising the appliances of people responsible for theft. Architectural models of smart meter, communication system, harmonic generator, and hybrid filter are proposed. Operational principle of the proposed system is illustrated in detail. For illustration, cost-benefit analysis for implementation and maintenance of the proposed system in India is presented.

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